

Purple: Combining Individual and Collective Action to Increase Online Sustainability

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Abstract

Technology has become embedded in everyday life, leading to a rapidly increasing online carbon footprint, yet individuals lack awareness of the environmental impact of these activities. We created the Purple app, targeting students as our users, to help them understand, monitor, and reduce their online carbon footprint in an engaging and practical way. We conducted user research to explore perceptions of their online carbon footprint and behaviours. Based on these insights, we iteratively designed and prototyped an app that integrates gamification, social accountability, and educational tools to encourage sustainable online habits. User testing highlighted the importance of clear information architecture, consistent metrics, and features that combine individual progress tracking with community-based motivation. This work contributes to the current HCI literature by addressing the environmental challenges associated with digital consumption. The Purple app serves as a model for how environmental awareness can be integrated into everyday technology use.

CCS Concepts

• Human-centered computing → Empirical studies in HCI.

Keywords

Interactive Technology, Tracking Technology, Online Carbon Footprint, Student Awareness, Greenhouse Gas Emission, Climate Change, Individual Impact

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1 Introduction

As the demand for consumer technology, artificial intelligence (AI) models, online streaming, and social networking platforms rises exponentially, so does the resulting greenhouse gas (GHG) emissions [12]. This increased demand has dramatically increased electricity consumption from data centres, servers, and network infrastructure, collectively referred to as information and communication technologies (ICT).

While this is a matter of immediate concern, it is met with a minimal sense of urgency amongst individuals due to a lack of awareness about the severity of an online carbon footprint. For example, many individuals are unaware that GHG emissions from ICT exceed even that of the aviation industry, an industry often criticised for its high carbon footprint [6, 12]. The digital carbon footprint is particularly alarming considering that global GHG emissions from ICT are projected to triple at minimum over the next decade [2, 8]. Equally important but often overlooked is the significant water footprint that cools these power-hungry hardware accelerators. By 2027, global AI demand could account for up to six times Denmark's annual water withdrawal or half that of the United Kingdom's [9]. Despite these alarming findings, individuals are largely unaware of the extent to which their personal online habits impact GHG emissions. Hence, we aim to address this information gap by advocating for increased awareness of the digital carbon footprint to meet the United Nation's Sustainable Development Goals [14], sustainable production and consumption (SDG 12) and climate action (SDG 13).

We focused our research on consumers' social responsibility to address the global carbon footprint rather than that of other major stakeholders such as technology corporations or institutional regulators. While corporations make up the bulk of GHG emissions globally, their main focus is increasing product efficiency to satisfy consumer demand rather than reducing GHG emissions [11]. However, they are now approaching a limit on how efficient ICT's can be, and the increasing efficiency has only led to higher consumer demand, creating a constant need for more energy [10]. In effect, the increasing digital carbon footprint of corporations is tied to the increasing consumer demand; therefore, we argue that it is

the responsibility of individual consumers to change their online consumption behaviours to address this issue.

1.1 Related work

One of the main issues of sustainable consumption is that individuals lack information about the carbon emissions resulting from their consumption choices [7]. This “intention behaviour gap” has been previously addressed via carbon footprint calculators [13] and carbon footprint tracking apps [5]. However, both approaches require individuals to self-estimate and manage their online behaviours which can be tedious and demotivating. We aim to build on this framework by tracking users’ online consumption and providing information that aligns with users’ daily activities, which has been shown to enhance information acquisition [15] and reduce carbon emissions by 23% [5].

2 User research

2.1 Online survey

We focused on undergraduate and postgraduate students as our user group because they are typically more invested in the global future and more willing to change their behaviour [1]. We conducted an online survey to understand students’ perceptions of online carbon footprints and their interest in reducing their impact. We received 34 responses after distributing the survey to students through popular social applications including WhatsApp and Reddit. After analysing our survey results, we discovered that participants not only perceived their screen time to be much lower than it actually was, but also lacked awareness of the environmental impact of everyday online activities (see Figure 1). Only 51% expressed interest in reducing their online carbon footprint.

From these results, we identified three areas to explore in our interviews: users’ understanding of their own consumption habits, their knowledge of how online activity contributes to climate change, and their interest in reducing their online carbon footprint. We specifically wanted to see if there was a relationship between users’ awareness of their carbon footprint and their interest in reducing it, as much of the existing literature identifies awareness as a principal motivator for behaviour change [4].

2.2 Semi-structured interviews

We conducted nine semi-structured interviews with survey respondents and used deductive thematic analysis to identify key findings. We discovered that most participants felt a personal responsibility to reduce their carbon footprint but were uncertain how to take meaningful action and whether it would be feasible for their lifestyle. Many expressed a general familiarity with the concept of an online carbon footprint, but found it difficult to conceptualise concretely and felt little pressure to change their behaviour. From this analysis, we determined that participants desire a product that is low time commitment, has high value, provides intuitive visualisations, and is personalised to the user. There was also a persistent sentiment that decreasing one’s carbon footprint is a shared responsibility amongst both individuals and larger entities such as celebrities and corporations, yet not everyone does their



Figure 1: Analysis of survey data

part. We realised that to build credibility with our users, our product must incorporate social accountability and address the largest contributors to climate change.

Furthermore, users expressed a desire for a solution that would not burden them or make them feel guilty for instances in which they felt technology was useful. One user who expressed low interest in reducing their carbon footprint felt that technology was too essential to their work in the STEM field to consider decreasing significantly. Additionally, several users distinguished between ‘productive’ and ‘time-wasting’ activities. While they were unwilling to reduce online usage for productive activities such as work, they mentioned wanting to reduce time-wasting activities (i.e. mindless scrolling on social media) for alternative reasons such as mental health and saving energy costs. According to this data, it is important to target specific online behaviours rather than generalised consumption to make the most effective impact amongst students. Consistent with our survey, several participants described themselves as ‘average users’ despite reporting screen times far above or below average. This inspired us to introduce an aspect of social awareness and peer influence to increase awareness of carbon footprint and motivation to adjust their usage.

2.3 Identified user requirements

Using the data from our interviews and survey, we identified several key user requirements:

- Be convenient and easy to integrate into daily life
- Visualise carbon emissions
- Use relatable metaphors to clearly understand impact
- Include social accountability
- Bring awareness to eco-friendly alternatives
- Personalise usage information to the user
- Avoid obstructing users from or inciting guilt for necessary use

3 Design process

After conducting interviews, we began brainstorming with a “Crazy 8’s” exercise to generate a wide range of ideas (see Figure 2). We shared and refined these ideas, often altering or combining them based on group feedback. Next, we created a user persona using data from our research to better visualise and empathise with our user (see Figure 3). During this process, we discussed how to align our user requirements and the requirements of the design brief with

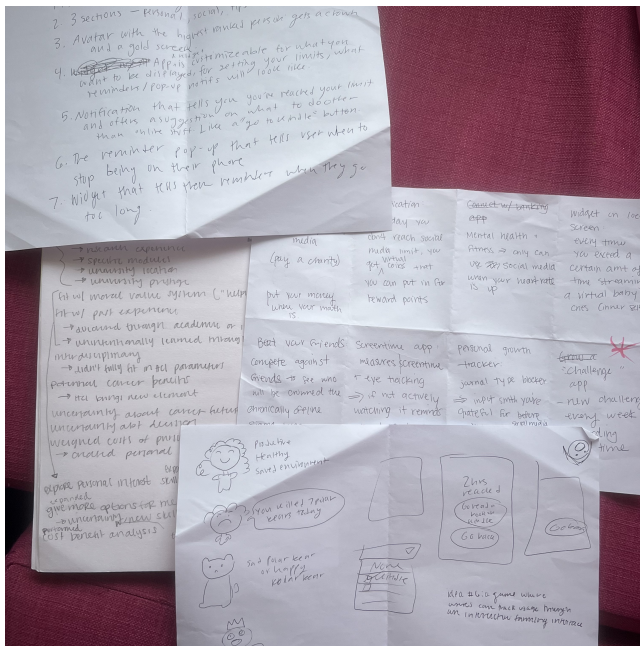


Figure 2: Crazy 8's design process

our users' current journey. This helped us develop the following selection criteria: **convenient, relatable, intuitive, engaging, and personalised**.

We used these selection criteria to evaluate our ideas, eventually narrowing them down to three options:

- (1) **Browser extension:** a downloadable extension that tracks user activity in the background and provides notifications for overuse, as well as a weekly summary.
- (2) **Home screen widget and app:** a widget that provides real-time user updates as well as further personalised user usage information in an application.
- (3) **Gamification app:** an application with a virtual world whose state mimics that of users' online carbon footprint (e.g., higher usage would lead to decaying plants and pollution or water saved on their phone would correlate with water to use in-app).

We then created a storyboard and journey map based on our survey and interview data (see Figure 4). We used our persona, journey map, and selection criteria to debate which product to pursue. The main trade-off we considered was balancing the delivery of important information in an engaging and impactful way while avoiding creating a technology that would further contribute to users' online carbon footprint.

To decide on a single product, each group member shared their favourite idea, highlighting the features they felt best satisfied our selection criteria. This process helped us identify which features met our user requirements. Based on group voting, we decided to pursue a mobile app product. However, through our evaluation process, we enhanced the idea by incorporating essential features from other ideas that we had set aside. For example, we considered a competition feature to integrate gamification and a home-screen

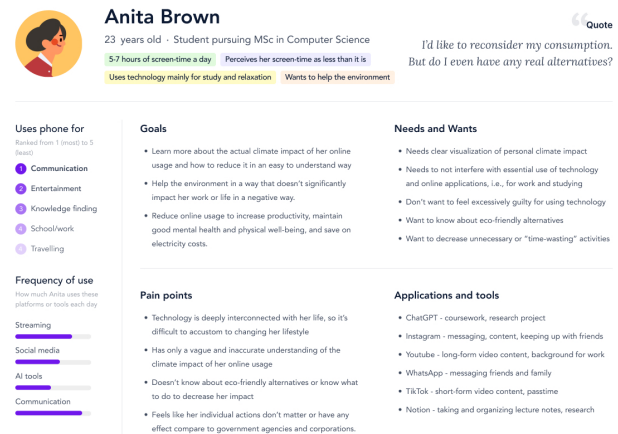


Figure 3: User persona depicting a standard user

widget to maintain convenience. We created several wireframes to illustrate and adjust our concept before settling on the core design features that best met our user requirements.



Figure 4: Storyboard depicting a user's morning routine in relation to technology

4 Mid-fidelity design

Using low- and mid-fidelity prototyping, we developed the first version of our product: the Purple app (see Figure 5). We drew inspiration from existing tracking apps such as Apple Health and Screen Time because we aimed to design a product that could be

seamlessly incorporated into users' daily lifestyles. Our product features a lock-screen widget component that provides the user with a quick visual of their wasteful phone usage. For more information, users can select the widget or directly open the app.

Throughout the app, usage was defined in familiar, tangible terms, such as using water and driving a car. We also included a feature that offered reminders or alternative suggestions for when users reached limits they set to reduce use. This feature directed users to more sustainable activities to help them change unwanted habits or behaviours. We considered the different wants and needs of novice, intermediate, and expert users. This led us to develop a range of functionalities for the feature – from simple, automatic options to customisable settings.

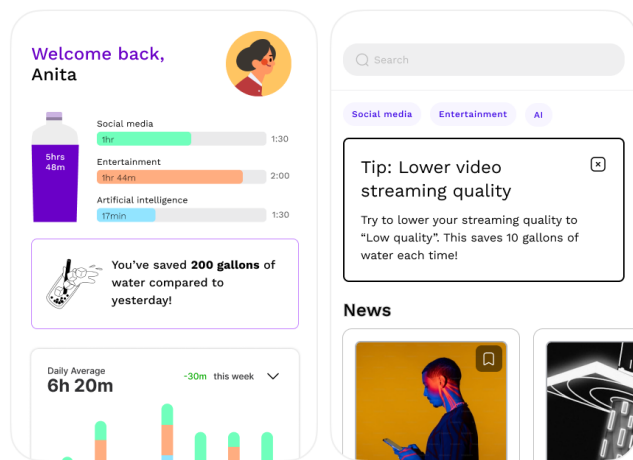


Figure 5: Sections of the mid-fidelity mockups of the home page (left) and the info page (right)

4.1 Pages

The initial low-fidelity version of the app was divided into two pages: Home and Information. The Home page functions as a user dashboard, breaking user usage into easy-to-interpret graphics. To narrow the scope of 'phone use' we considered both high carbon-emitting activities and those that users wanted to limit. This, along with our background research, led us to focus on the categories of streaming, social media, and AI use [3, 9]. In the first version of the design, users' screen time in each category was displayed next to a graphic of a water bottle, whose water increased in conjunction with user activity to represent the amount of water being used up by the user's online consumption habits. The app also allowed users to set personal goals for each category, compensating for the wide range of online behaviour we documented in our user research. Further down the Home page was a leaderboard in which users could compare their screen time to that of their friends, family, and colleagues. This introduced the component of social responsibility brought up in user research and encouraged external accountability and personal investment.

We designed the Information page to provide users with detailed information on the online carbon footprint, focusing on topics they

were interested in. It provided a comprehensive understanding of what an online footprint is, how it is created, and how it can be reduced. It achieved this by suggesting practical tips to mitigate users' impact, sharing relevant news articles, and presenting petitions for users to support. This addressed a key user insight to incorporate actionable items, helping to make the app more positive and alleviating climate anxiety.

5 User testing

We recruited five students with a range of technical backgrounds to provide feedback on our initial prototype. We first conducted a think-aloud study, letting the user naturally explore the application to discover if they wanted to use the features we created. We then had users run through four tasks to test specific features. Each session took approximately 30 minutes.

5.1 User feedback

Improve information architecture: While users enjoyed the gamification aspect of the app, they found the leaderboard confusing. They had difficulty interpreting information from the feature because of its layout and oversimplification. They also felt that the transition between their personal energy usage to the community comparison was jarring and described the page as information overload.

Use consistent measurement metrics: While users found screen time an intuitive measurement, they struggled to interpret the relationship between carbon emissions and water usage as metrics to visualise their carbon footprint. They found the relatable examples unclear how screen time translated into the production of water and greenhouse gases.

Differentiate the impact of various online activities: Building on the feedback above, users expressed confusion about how overall screen time correlated to their online carbon footprint. They noted that different online activities, such as using generative AI versus streaming YouTube, produced very different levels of carbon emissions.

Give clear justification for technical decisions: Users expressed a desire for more comprehensive information about the app's background and how metrics were chosen, how carbon footprint is calculated, and our sources.

Enjoyed petition feature: Lastly, users enjoyed the petition aspect of the app. They expressed that the opportunity to engage in collective efforts to address climate change made the app feel more impactful.

5.2 Design changes

To address the user feedback, we executed some key changes: we clarified our metrics system, implemented a monthly challenge feature, and created a new "Friends" page and "Details" page (see Table 1).

We brainstormed alternatives to presenting solely screen time, including one user's suggestion to create a points system by calculating a weighted average based on carbon emissions and water used for each activity. However, we felt that requiring users to learn a new system for our app would go against a key user requirement: convenience. We generated ideas for what metrics to use and created a C-Box to compare how easy-to-understand each metric was with how impactful it would be (see Figure 6). This led us to make our primary metric carbon footprint measured in greenhouse gas

emissions, and our secondary metric as screen time. We kept the relatable examples users enjoyed but presented them as supplementary details. We included visual cues for users to easily interpret their footprint at a glance (e.g., using red for high emissions).

To emphasise the positive impact of collective action, we introduced monthly challenges. This encourages individual action by setting online sustainability challenges app-wide, and displays the number of participating users and the total carbon savings achieved. Lastly, we decided to incorporate more information about users' personal progress on their home screen, including real-world visualisations as well as a trends chart. These adjustments aim to remind the user of their individual impact and encourage continued engagement with their sustainability goals.

Moving social features to a new tab reduced the overlap between functions, facilitated easier app navigation, and improved the overall user experience. We also created a "Details" page that explains the equation and source behind each calculation. The menu includes an "About" page that provides an explanation of the app's mission to reduce online carbon footprints. Our final design incorporates these user-driven modifications (see Figure 7).

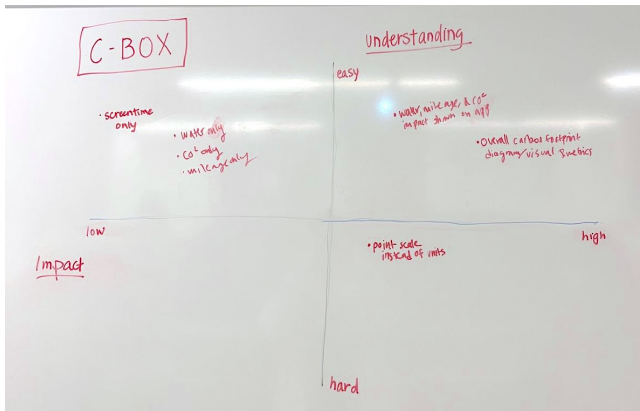


Figure 6: C-box analysis of user testing

6 Discussion

The Purple app combines gamification and education to raise awareness about the environmental impact of online activities and promote sustainable actions. By focusing on both community and individual responsibility towards sustainability, we aim to empower users to reduce their online carbon footprint.

Our main challenge during this user centred design (UCD) process was balancing the effectiveness with user experience. Unlike goal-oriented apps like "Couch to 5K" or Apple's fitness rings that encourage users to increase activity, our end goal is to reduce online activity and lower online carbon footprint. The challenge is how to motivate users to decrease their online activity without eliciting negative feelings, like guilt or demoralisation. We addressed this by incorporating a community-driven competition structure, which encourages positive reinforcement and helps users stay motivated to reduce their online carbon footprint.

Due to the wealth of detailed information generated, user interviews were essential in informing our design and ideation process.

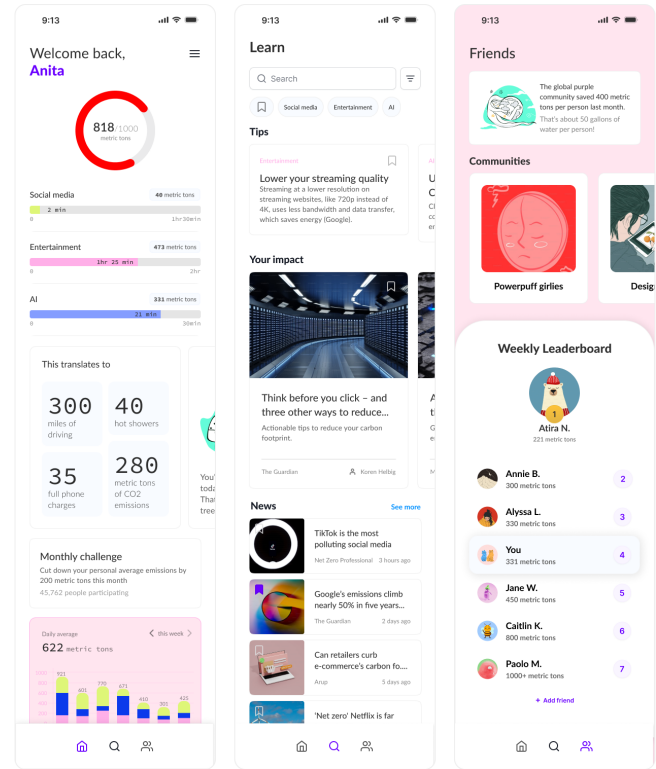


Figure 7: Final design for the "Home" page (left) the "Learn" page, and the "Friends" page (right). Users can access these pages via the bottom navigation bar.

The user testing sessions further clarified what aspects of our design to modify for increased impact and better usability. Initially, we focused solely on screentime for simplicity, but feedback led us to include the corresponding carbon emissions. This directly connects online activity with carbon footprint, making the data more tangible and actionable for users. We also recognised that users are unwilling to drastically modify their current routines. Based on user feedback, we focused on phone usage since users reported often "wasting" time on their phones, while they used computers for coursework and other unavoidable work tasks. While we could potentially expand to the other ideas we explored during our ideation process, such as browser extensions for computers or a gamification application, we ultimately decided against those for several reasons. Specifically, a computer-based browser extension would mainly affect essential work-related tasks, and a highly gamified app might be overly engaging, which could lead users to spend more time online and inadvertently increase their online carbon footprint. Our design approach reflects a balance of individual, community, and societal responsibility for online sustainability. Through both individual and community action, users can learn about the environmental costs of their digital behaviours and take tangible steps to reduce their carbon footprint. Overall, the Purple app provides users with practical tools to fight against the climate crisis.

User Feedback	Design Changes
Improve information architecture	Create a new page for social features Add numbering to leaderboard Label different communities clearly
Provide positive feedback	Include the amount of carbon emissions user has saved Create monthly app-wide challenges Add 'amount of carbon emissions saved' to reminder feature
Use consistent measurement metric	Change from measuring water usage in gallons and gas emissions in miles driven to carbon footprint in metric tons
Make screen time bars more usable	Increase size of bars Minimise info presented (with further details available through interaction with the feature)
Explain the justification for our metrics more clearly	Make technical justification accessible for each calculation Add a general information page
Enjoyed petition feature	Consider personal recommendations based on user activity and updates on petitions they have signed

Table 1: User Feedback and Design Changes

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